# Lecture 4

CSS3223 Calculus



## CHAPTER 2 :
## LIMITS AND CONTINUITY

2.1 RATES OF CHANGE AND TANGENTS TO CURVES
2.2 LIMIT OF A FUNCTION AND LIMIT LAWS
2.3ONE-SIDED LIMITS AND LIMITS AT INFINITY
2.4 INFINITE LIMITS AND VERTICAL ASYMPTOTES
2.5 CONTINUITY
2.6 TANGENTS AND DERIVATIVES AT A POINT

# Recollection



- We have seen:
  - Limits arise when finding the instantaneous rate of change of a function or the tangent to a curve
  - Informal definition of limit
  - Precise definition of limit
  - How to calculate values of limit
  - Some limit laws
  - Sandwich theorem

# One-Sided Limits and Limits at Infinity



- Objectives:
  - To extend the limit concept to one-sided limits
    - i.e. limits as $x$ approaches the number $a$ from the left-hand side (where $x < a$) or the right-hand side ($x > a$) only.
  - To analyze the graphs of certain rational functions as well as other functions with limit behavior as $x \to \pm\infty$

# One-Sided Limits



- If $f$ fails to have a two-sided limit at $c$, it may still have a one-sided limit i.e. a limit if the approach is from one side only.

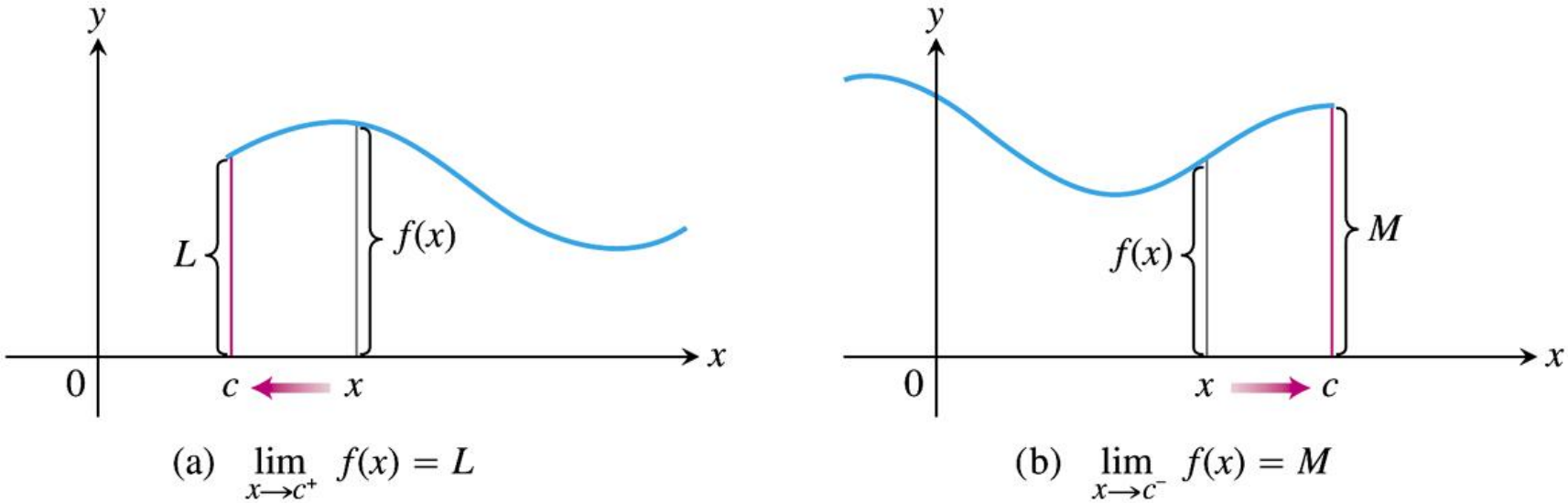



**FIGURE 2.25** (a) Right-hand limit as $x$ approaches $c$. (b) Left-hand limit as $x$ approaches $c$.

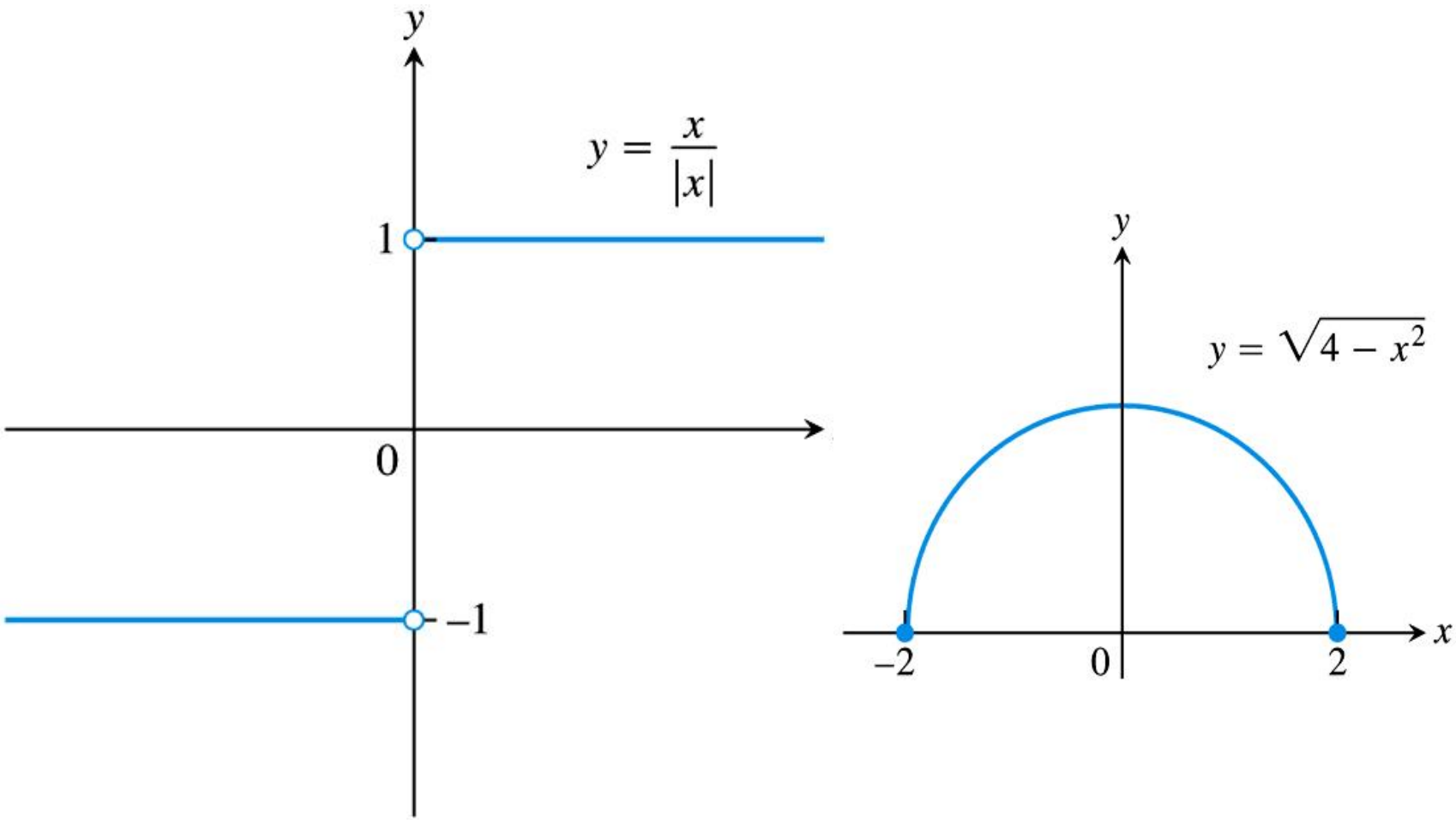



**FIGURE 2.24** Different right-hand and left-hand limits at the origin.

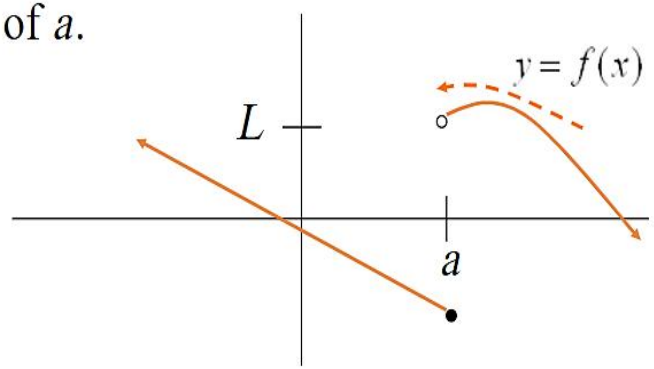
# One-Sided Limit



The right-hand limit of $f(x)$, as $x$ approaches $a$, equals $L$

written: $\lim_{x \to a^+} f(x) = L$

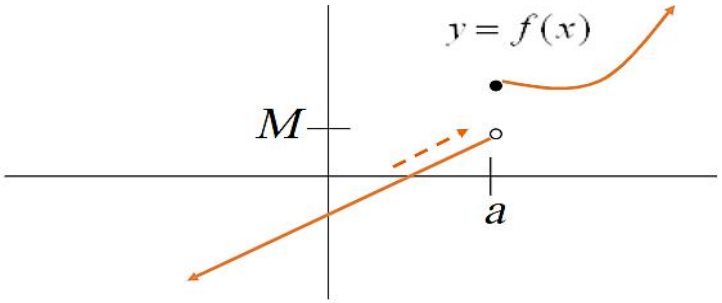
if we can make the value $f(x)$ arbitrarily close to $L$ by taking $x$ to be sufficiently close to the right of $a$.




The left-hand limit of $f(x)$, as $x$ approaches $a$, equals $M$

written: $\lim_{x \to a^-} f(x) = M$

if we can make the value $f(x)$ arbitrarily close to $M$ by taking $x$ to be sufficiently close to the left of $a$.



# Examples



1. Given $f(x)=\begin{cases} x^2 & \text{if } x \le 3 \\ 2x & \text{if } x > 3 \end{cases}$

Find $\lim\limits_{x \to 3^+} f(x)$

Find $\lim\limits_{x \to 3^-} f(x)$

Ans:

$$\boxed{\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} 2x = 6}$$

$$\boxed{\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} x^2 = 9}$$

2. Let $f(x)=\begin{cases} x+1, & \text{if } x>0 \\ x-1, & \text{if } x\le 0. \end{cases}$ Find the limits:

a) $\lim\limits_{x\to 0^+} f(x)$

b) $\lim\limits_{x\to 0^-} f(x)$

c) $\lim\limits_{x\to 1^-} f(x)$

d) $\lim\limits_{x\to 1^+} f(x)$

Ans:

a) $\lim\limits_{x\to 0^+} f(x)=\lim\limits_{x\to 0^+}(x+1)=0+1=1$

b) $\lim\limits_{x\to 0^-} f(x)=\lim\limits_{x\to 0^-}(x-1)=0-1=-1$

c) $\lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^-}(x+1)=1+1=2$

d) $\lim\limits_{x\to 1^+} f(x)=\lim\limits_{x\to 1^+}(x+1)=1+1=2$

# Two-Sided Limits



- To have a limit $L$ as $x$ approaches $c$, a function $f$ must be defined on both sides of $c$ and its values $f(x)$ must approach $L$ as $x$ approaches $c$ from either side. Because of this, ordinary limits are called **two-sided** limits.

# Theorem



**THEOREM 6**

A function $f(x)$ has a limit as $x$ approaches $c$ if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x \to c} f(x) = L \quad \Leftrightarrow \quad \lim_{x \to c^-} f(x) = L \quad \text{and} \quad \lim_{x \to c^+} f(x) = L.$$

This theorem is used to show a limit does not exist.

eg. For the function $f(x) = \begin{cases} x+1, & \text{if } x > 0 \\ x-1, & \text{if } x \leq 0 \end{cases}$

$\lim_{x \to 0} f(x)$ does not exist because $\lim_{x \to 0^+} f(x) = 1$ & $\lim_{x \to 0^-} f(x) = -1$.

But, $\lim_{x \to 1} f(x) = 2$ because $\lim_{x \to 1^+} f(x) = 2$ & $\lim_{x \to 1^-} f(x) = 2$.

# Example



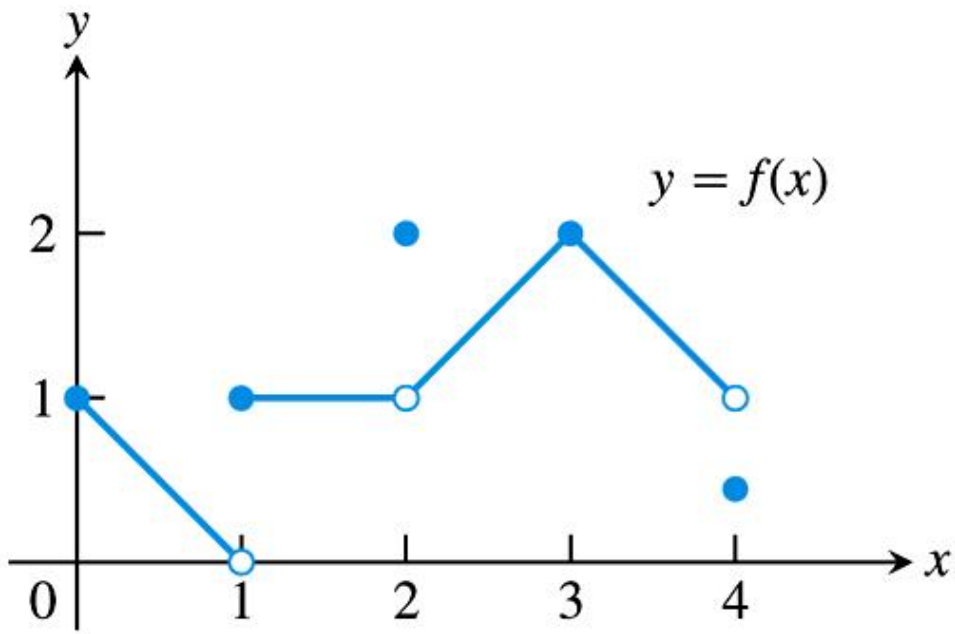



**FIGURE 2.27** Graph of the function in Example 2.

Identify the two-sides limits at:

$x=0$, $x=1$, $x=2$, $x=3$
$x=4$

# Limit Theorems



If $c$ is any number, $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$, then

a) $\lim_{x \to a} \left( f(x) + g(x) \right) = L + M$

b) $\lim_{x \to a} \left( f(x) - g(x) \right) = L - M$

c) $\lim_{x \to a} \left( f(x) \cdot g(x) \right) = L \cdot M$

d) $\lim_{x \to a} \left( f(x) / g(x) \right) = L / M, \ (M \neq 0)$

e) $\lim_{x \to a} \left( c \cdot f(x) \right) = c \cdot L$

f) $\lim_{x \to a} \left( f(x) \right)^n = L^n$

g) $\lim_{x \to a} c = c$

h) $\lim_{x \to a} x = a$

i) $\lim_{x \to a} x^n = a^n$

j) $\lim_{x \to a} \sqrt{f(x)} = \sqrt{L}, \ (L > 0)$

# Examples using Limit Rules



Ex1: $\lim_{x\to 3}\left(x^2+1\right)$

$$=\lim_{x\to 3}x^2+\lim_{x\to 3}1$$

$$=\left(\lim_{x\to 3}x\right)^2+\lim_{x\to 3}1$$

$$=3^2+1=10$$

Ex2: $\lim_{x\to 1}\frac{\sqrt{2x-1}}{3x+5}$

$$=\frac{\sqrt{\lim_{x\to 1}\left(2x-1\right)}}{\lim_{x\to 1}\left(3x+5\right)}$$

$$=\frac{\sqrt{2\lim_{x\to 1}x-\lim_{x\to 1}1}}{3\lim_{x\to 1}x+\lim_{x\to 1}5}$$

$$=\frac{\sqrt{2-1}}{3+5}=\frac{1}{8}$$

# Examples using Limit Rules



Suppose $\lim_{x \to 3} f(x) = 4$ and $\lim_{x \to 3} g(x) = -2$. Find

Ex3: $\lim_{x \to 3} \left( f(x) + g(x) \right)$

Ex4: $\lim_{x \to 3} \left( f(x) - g(x) \right)$

Ans:

$$\lim_{x \to 3} \left( f(x) + g(x) \right)$$

$$= \lim_{x \to 3} f(x) + \lim_{x \to 3} g(x)$$

$$= 4 + (-2) = 2$$

$$\lim_{x \to 3} \left( f(x) - g(x) \right)$$

$$= \lim_{x \to 3} f(x) - \lim_{x \to 3} g(x)$$

$$= 4 - (-2) = 6$$

Ex5: $\lim_{x \to 3} \left( \frac{2f(x) - g(x)}{f(x)g(x)} \right)$

$$= \frac{\lim_{x \to 3} 2f(x) - \lim_{x \to 3} g(x)}{\lim_{x \to 3} f(x) \cdot \lim_{x \to 3} g(x)}$$

$$= \frac{2 \cdot 4 - (-2)}{4 \cdot (-2)}$$

$$= \frac{-5}{4}$$

# Indeterminate Forms



- Indeterminate forms occur when substitution in the limit results in 0/0. In such cases either factor or rationalize the expressions.

Ex1.

$$\lim_{x \to -5} \frac{x+5}{x^2-25}$$

Notice $\frac{0}{0}$ form

$$= \lim_{x \to -5} \frac{x+5}{(x-5)(x+5)}$$

Factor and cancel common factors

$$= \lim_{x \to -5} \frac{1}{(x-5)} = \frac{1}{-10}$$

Ex2.

$$\lim_{x\to 9}\left(\frac{\sqrt{x}-3}{x-9}\right) = \lim_{x\to 9}\left(\frac{(\sqrt{x}-3)(\sqrt{x}+3)}{(x-9)(\sqrt{x}+3)}\right)$$

$$= \lim_{x\to 9}\left(\frac{x-9}{(x-9)(\sqrt{x}+3)}\right) = \lim_{x\to 9}\left(\frac{1}{\sqrt{x}+3}\right) = \frac{1}{6}$$

Ex3.

$$\lim_{x\to -2}\left(\frac{4-x^2}{2x^2+x^3}\right) = \lim_{x\to -2}\left(\frac{(2-x)(2+x)}{x^2(2+x)}\right) = \lim_{x\to -2}\left(\frac{2-x}{x^2}\right)$$

$$= \frac{2-(-2)}{(-2)^2} = \frac{4}{4} = 1$$

# Limits involving (sin θ)/ θ



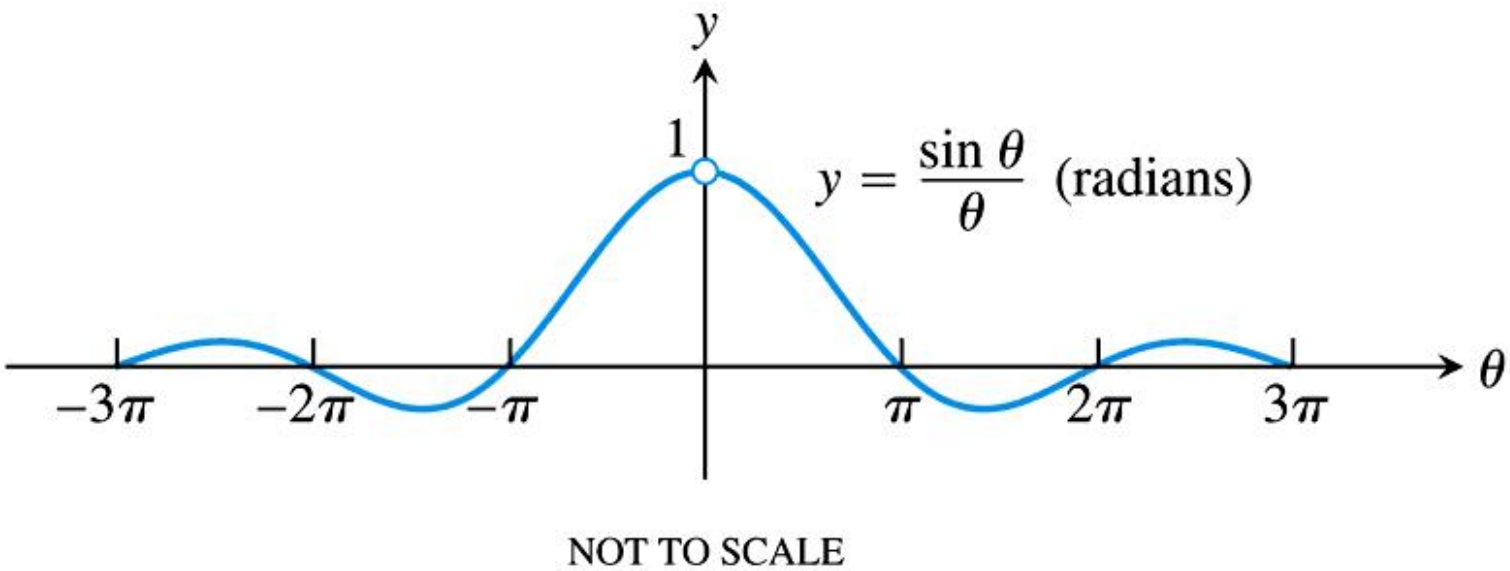



**FIGURE 2.32** The graph of $f(\theta) = (\sin\theta)/\theta$ shows that the right- and left-hand limits as $\theta$ approaches 0 are both 1.

# Theorem



**THEOREM 7**

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1 \qquad (\theta \text{ in radians}) \tag{1}$$

# Limits at Infinity (as $x \to \pm\infty$)



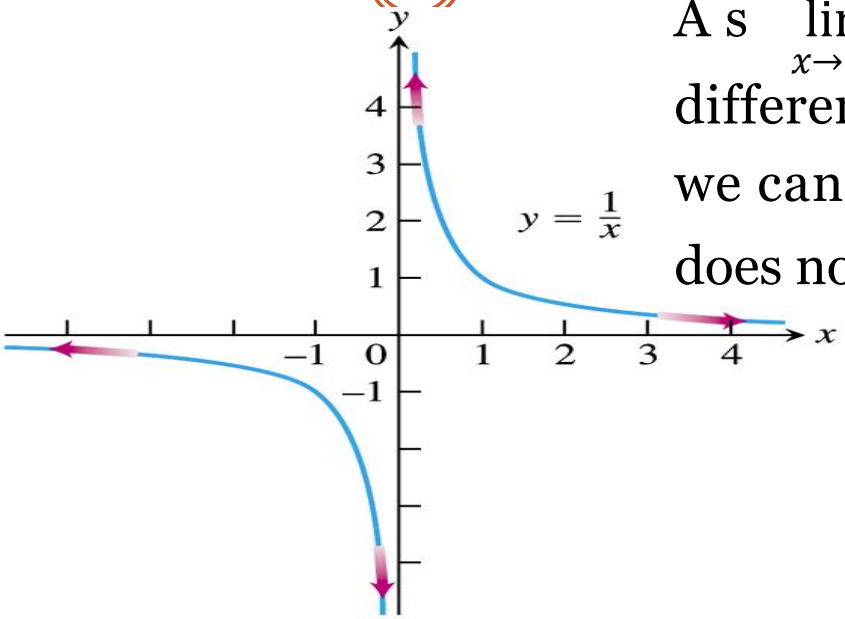



**FIGURE 2.34** The graph of $y = 1/x$ approaches 0 as $x \to \infty$ or $x \to -\infty$.

As $\lim_{x\to 0^+} \frac{1}{x}$ & $\lim_{x\to 0^-} \frac{1}{x}$ have different values, therefore, we can conclude that $\lim_{x\to 0} \frac{1}{x}$ does not exist!

$\lim_{x \to 0^+} \frac{1}{x} = \infty$, $\lim_{x \to 0^+} \frac{1}{x}$ does not exist because $\frac{1}{x}$ becomes arbitrarily large and positive as $x \to 0^+$.

$\lim_{x \to 0^-} \frac{1}{x} = -\infty$, $\lim_{x \to 0^-} \frac{1}{x}$ does not exist because $\frac{1}{x}$ becomes arbitrarily large and negative as $x \to 0^-$.

## DEFINITIONS

**1.** We say that $f(x)$ has the **limit $L$ as $x$ approaches infinity** and write

$$\lim_{x \to \infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $M$ such that for all $x$

$$x > M \qquad \Rightarrow \qquad |f(x) - L| < \epsilon.$$

**2.** We say that $f(x)$ has the **limit $L$ as $x$ approaches minus infinity** and write

$$\lim_{x \to -\infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $N$ such that for all $x$

$$x < N \qquad \Rightarrow \qquad |f(x) - L| < \epsilon.$$

## THEOREM 8 Limit Laws as $x \to \pm\infty$

If $L$, $M$, and $k$, are real numbers and

$$\lim_{x\to\pm\infty} f(x) = L \quad \text{and} \quad \lim_{x\to\pm\infty} g(x) = M, \quad \text{then}$$

1. *Sum Rule:* $\lim_{x\to\pm\infty} (f(x) + g(x)) = L + M$
2. *Difference Rule:* $\lim_{x\to\pm\infty} (f(x) - g(x)) = L - M$
3. *Product Rule:* $\lim_{x\to\pm\infty} (f(x) \cdot g(x)) = L \cdot M$
4. *Constant Multiple Rule:* $\lim_{x\to\pm\infty} (k \cdot f(x)) = k \cdot L$
5. *Quotient Rule:* $\lim_{x\to\pm\infty} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$
6. *Power Rule:* If $r$ and $s$ are integers with no common factors, $s \neq 0$, then

$$\lim_{x\to\pm\infty} (f(x))^{r/s} = L^{r/s}$$

provided that $L^{r/s}$ is a real number. (If $s$ is even, we assume that $L > 0$.)

# Examples



- Find

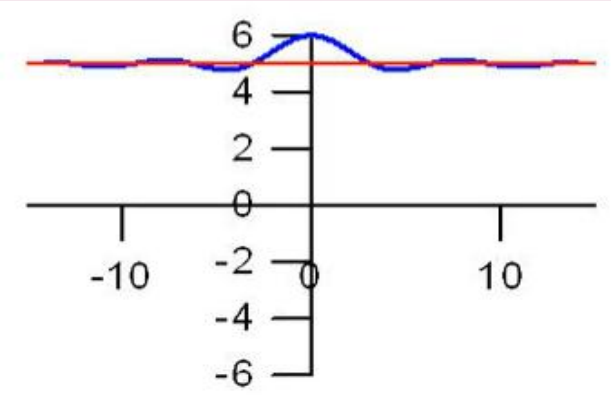
a) $$\lim_{x \to \infty}\left(5 + \frac{1}{x}\right) = 5$$

b) $$\lim_{x \to \infty} \frac{\pi\sqrt{3}}{x^2} = 0$$

c) Find $\lim_{x\to\infty} \frac{5x+\sin x}{x}$

$$\frac{5x+\sin x}{x} = \frac{5x}{x} + \frac{\sin x}{x} = 5 + \frac{\sin x}{x}$$

$$\lim_{x\to\infty} \frac{5x+\sin x}{x} = \lim_{x\to\infty} 5 + \lim_{x\to\infty} \frac{\sin x}{x} = 5$$



# Limits at Infinity of Rational Functions



- To determine the limit of a rational function as $x \to \pm\infty$, we can divide the numerator and denominator by the **highest power of $x$ in the denominator.**
- Example:

a) $$\lim_{x\to\infty}\frac{5x^2+8x-3}{3x^2+2}=\lim_{x\to\infty}\frac{5+\frac{8}{x}-\frac{3}{x^2}}{3+\frac{2}{x^2}}=\frac{5}{3}$$

b) $$\lim_{x\to\infty}\frac{11x+2}{2x^3-1}=\lim_{x\to\infty}\frac{\frac{11}{x^2}+\frac{2}{x^3}}{2-\frac{1}{x^3}}=\frac{0}{2}=0$$

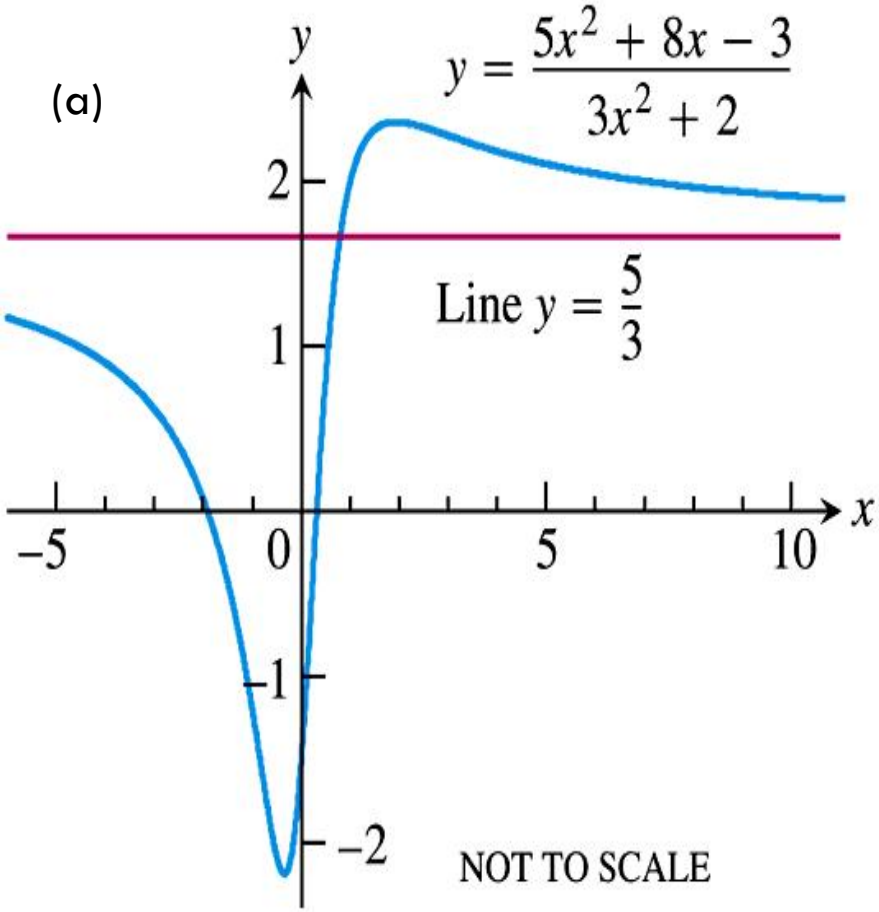



**FIGURE 2.36** The graph of the function in Example 8. The graph approaches the line $y = 5/3$ as $|x|$ increases.

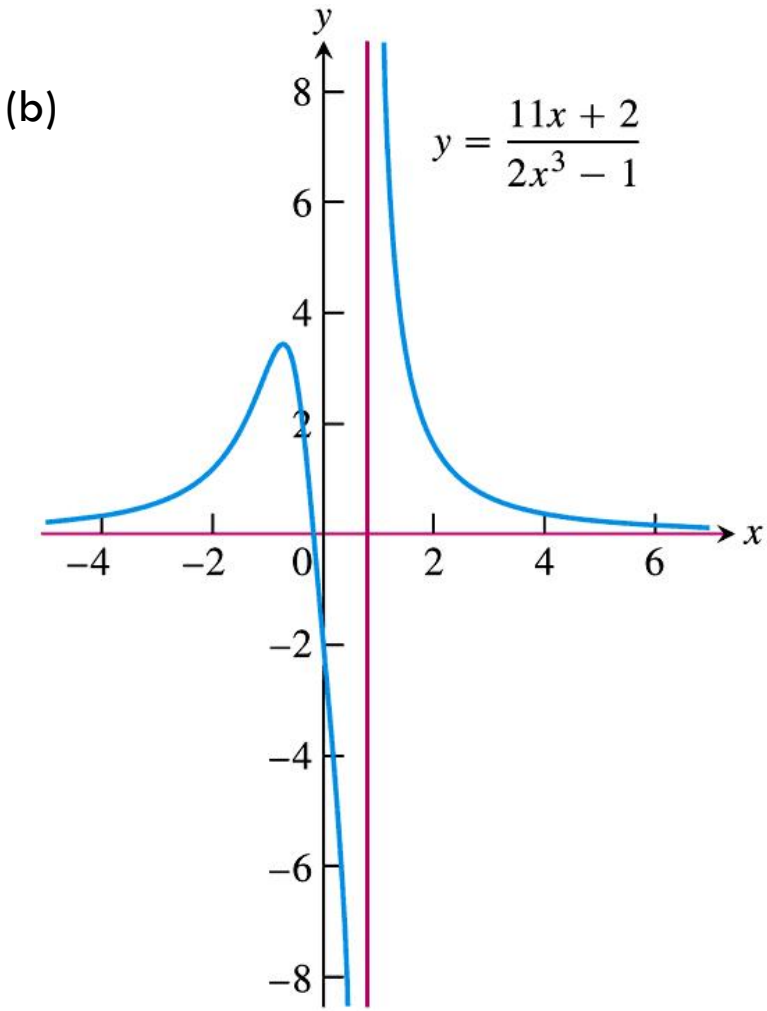



**FIGURE 2.37** The graph of the function in Example 9. The graph approaches the $x$-axis as $|x|$ increases.

c)

$$\lim_{x\to\infty}\left(\frac{4x^2-5x+21}{7x^3+5x^2-10x+1}\right)$$

$$=\lim_{x\to\infty}\left(\frac{\frac{4x^2}{x^3}-\frac{5x}{x^3}+\frac{21}{x^3}}{\frac{7x^3}{x^3}+\frac{5x^2}{x^3}-\frac{10x}{x^3}+\frac{1}{x^3}}\right)$$

$$=\lim_{x\to\infty}\left(\frac{\frac{4}{x}-\frac{5}{x^2}+\frac{21}{x^3}}{7+\frac{5}{x}-\frac{10}{x^2}+\frac{1}{x^3}}\right)$$

$$=\frac{0}{7}$$

$$=0$$

d)

$$\lim_{x\to\infty}\left(\frac{x^2+2x-4}{12x+31}\right)$$

$$=\lim_{x\to\infty}\left(\frac{\frac{x^2}{x}+\frac{2x}{x}-\frac{4}{x}}{\frac{12x}{x}+\frac{31}{x}}\right)$$

$$=\lim_{x\to\infty}\left(\frac{x+2-\frac{4}{x}}{12+\frac{31}{x}}\right)$$

$$=\frac{\infty+2}{12}$$

$$=\infty$$

# Horizontal Asymptotes



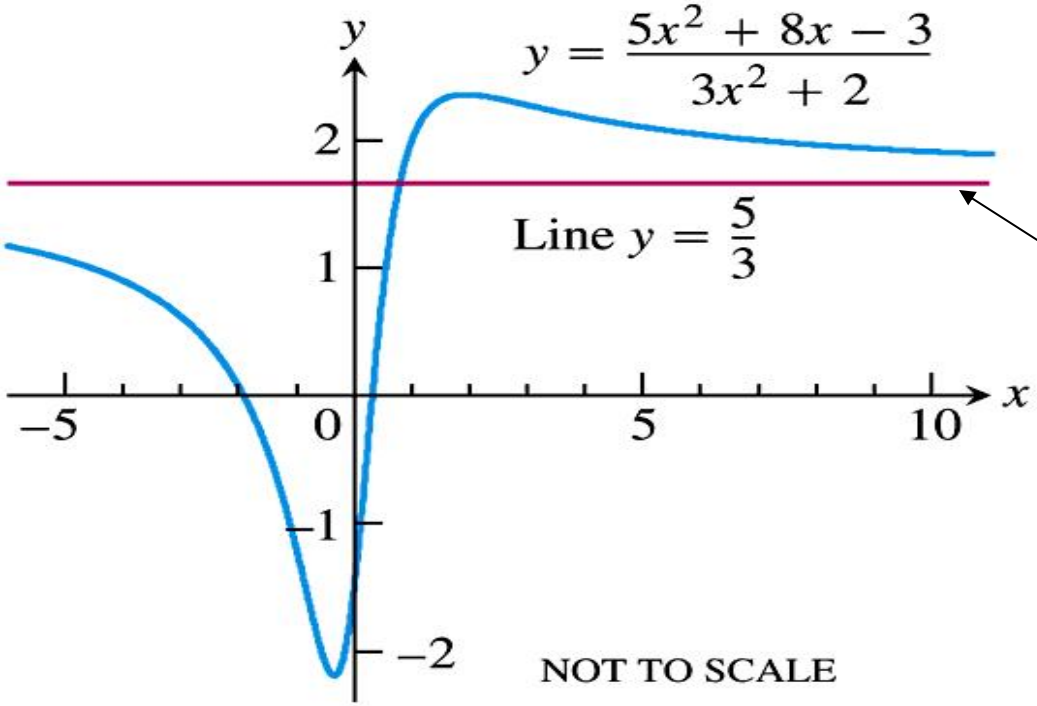
- If the distance between the graph of a function and some fixed line approaches zero as a point on the graph moves increasingly far from the origin, we say that the graph approaches the line asymptotically and the line is an *asymptote* of the graph.

**DEFINITION** A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \to \infty} f(x) = b \qquad \text{or} \qquad \lim_{x \to -\infty} f(x) = b.$$

# Example






Horizontal asymptote

$$\lim_{x \to \infty} f(x) = \frac{5}{3}$$

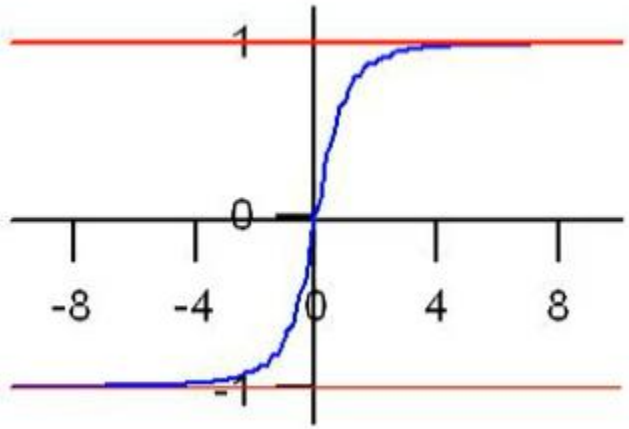
$$\lim_{x \to -\infty} f(x) = \frac{5}{3}$$

**FIGURE 2.36** The graph of the function in Example 8. The graph approaches the line $y = 5/3$ as $|x|$ increases.

# Example



Investigate the graph of $f(x) = \dfrac{x}{\sqrt{x^2+1}}$




From the graph, $\lim\limits_{x\to\infty}\left(\dfrac{x}{\sqrt{x^2+1}}\right) = 1$ & $\lim\limits_{x\to-\infty}\left(\dfrac{x}{\sqrt{x^2+1}}\right) = -1$

The function has 2 horizontal asymptotes: y = 1 & y = -1.

# Infinite limits and vertical asymptotes



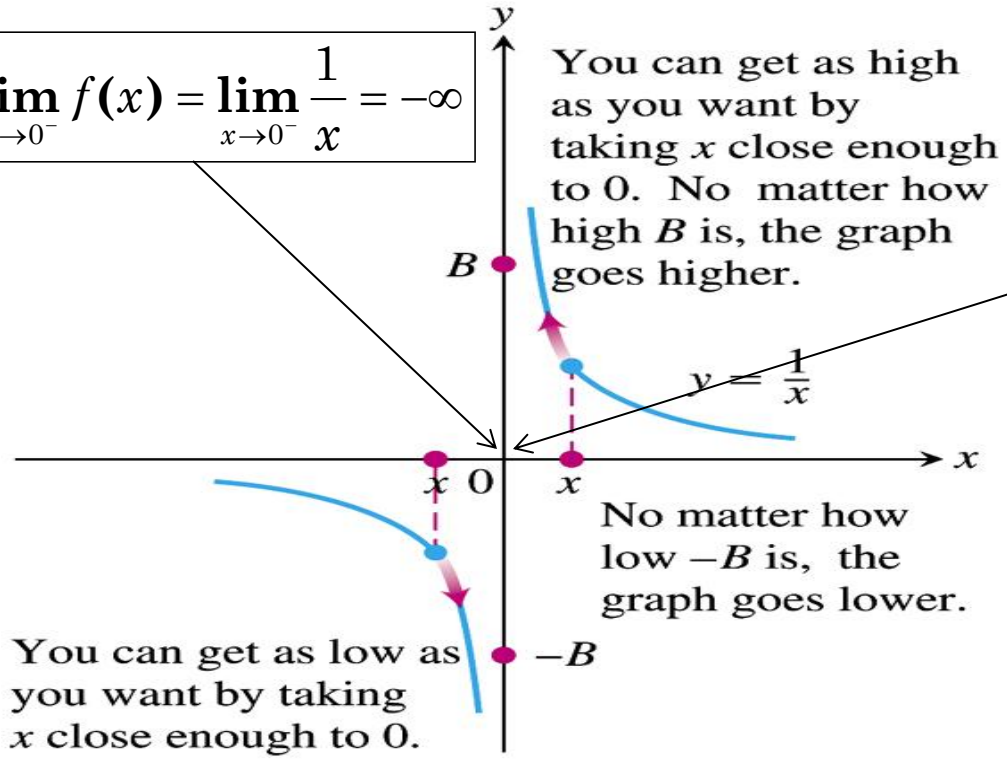
- Objectives:
  - To describe the behavior of functions whose values become arbitrarily large
  - To analyze graphs of rational functions using vertical asymptotes for numerically large values of $x$

# Infinite limits



$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{x} = \infty$$



# Infinite limits



- We say that the limit does not exist because the function becomes arbitrarily large and positive.

$$\boxed{\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{x} = \infty}$$

- We say that the limit does not exist because its values becomes arbitrarily large and negative.

$$\boxed{\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{x} = -\infty}$$

# Examples



Find

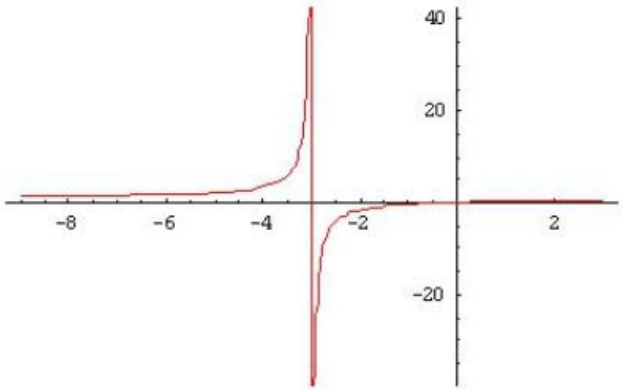
$$1.\lim_{x\to 2}\frac{x-2}{x^2-4}=\lim_{x\to 2}\frac{(x-2)}{(x-2)(x+2)}=\lim_{x\to 2}\frac{1}{(x+2)}=\frac{1}{4}$$

$$2.\lim_{x\to 2}\frac{x-3}{x^2-4}=\lim_{x\to 2}\frac{x-3}{(x-2)(x+2)}\text{ does not exist.}$$

Find the limits

3. $$\lim_{x\to 0^+}\left(\frac{3x^2+2x+1}{2x^2}\right)=\lim_{x\to 0^+}\left(\frac{3+2/x+1/x^2}{2}\right)=\frac{3+\infty+\infty}{2}=\infty$$

4. $$\lim_{x\to -3^+}\left(\frac{2x+1}{2x+6}\right)=\lim_{x\to -3^+}\left(\frac{2x+1}{2(x+3)}\right)=-\infty$$



# Precise definitions of infinite limits



**DEFINITIONS**

1. We say that **$f(x)$ approaches infinity as $x$ approaches $x_0$**, and write

$$\lim_{x \to x_0} f(x) = \infty,$$

if for every positive real number $B$ there exists a corresponding $\delta > 0$ such that for all $x$

$$0 < |x - x_0| < \delta \qquad \Rightarrow \qquad f(x) > B.$$

2. We say that **$f(x)$ approaches negative infinity as $x$ approaches $x_0$**, and write

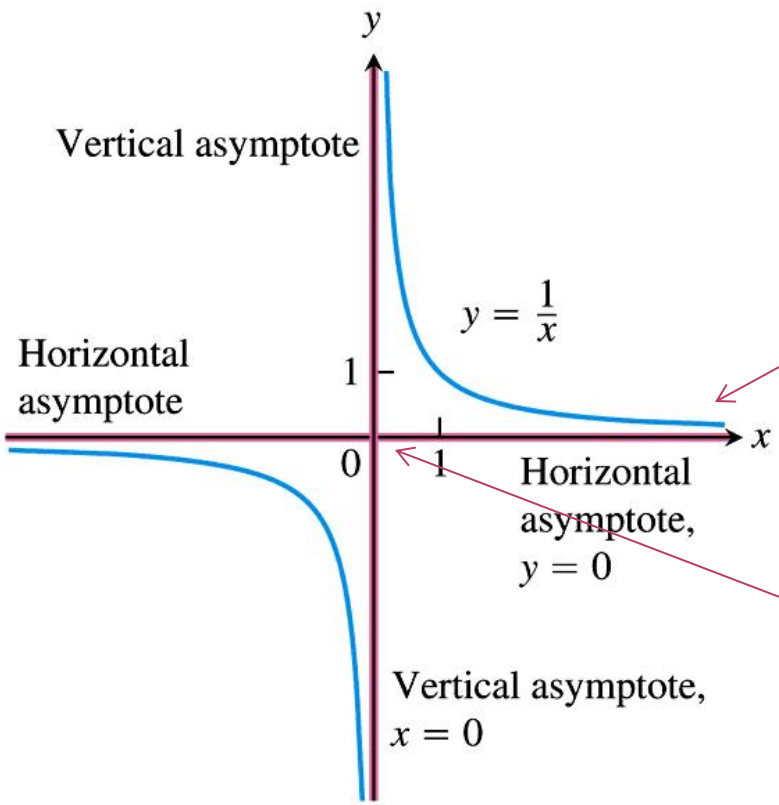
$$\lim_{x \to x_0} f(x) = -\infty,$$

if for every negative real number $-B$ there exists a corresponding $\delta > 0$ such that for all $x$

$$0 < |x - x_0| < \delta \qquad \Rightarrow \qquad f(x) < -B.$$

# Vertical asymptotes






$$\lim_{x \to +\infty} f(x) = 0$$

$$\lim_{x \to 0^+} f(x) = -\infty$$

**FIGURE 2.47** The coordinate axes are asymptotes of both branches of the hyperbola $y = 1/x$.

Observe that the denominator is zero at $x = 0$ and the function is undefined there.

**DEFINITION** A line $x = a$ is a **vertical asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \to a^+} f(x) = \pm\infty \qquad \text{or} \qquad \lim_{x \to a^-} f(x) = \pm\infty .$$

Find the asymptotes of the graphs of the functions

. $f(x) = \dfrac{x^2+1}{x^2-1}$

(i) $\lim\limits_{x \to 1^-} f(x) = -\infty$

Therefore the line $x = 1$ is a vertical asymptote.

(ii) $\lim\limits_{x \to -1^-} f(x) = +\infty.$

Therefore the line $x = -1$ is a vertical asymptote.

# Finding the horizontal asymptotes



- For rational functions:
  - If both polynomials are the same degree, divide the coefficients of the highest degree terms.

e.g., $$f(x) = \frac{6x^2 - 3x + 4}{2x^2 - 8}$$

both polynomials are 2nd degree, so the asymptote is at y=6/2=3.

  - If the polynomial in the numerator is a lower degree than the denominator, the x-axis (y = 0) is the horizontal asymptote.
  - If the polynomial in the numerator is a higher degree than the denominator, there is **no** horizontal asymptote.

# Finding the vertical asymptotes



- The vertical asymptotes come from the zeroes of the denominator, so, set the denominator equal to zero and solve! If the denominator has no zeroes, then there are no vertical asymptotes.

e.g.,

$$y = \frac{x^2 + 3x + 1}{4x^2 - 9}$$

$$4x^2 - 9 = 0,$$

$$4x^2 = 9,$$

$$x^2 = \frac{9}{4} \Rightarrow x = \pm\frac{3}{2}$$

So, the two vertical asymptotes are at $x = \pm\frac{3}{2}$

# Examples



Find the horizontal and vertical asymptotes of the curve

$$y = \frac{x+3}{x+2}$$

- Horizontal asymptote: 1/1=1.
- Vertical asymptote: $x + 2 = 0$, $x = -2$.

# Examples



- Find the asymptotes of the function

$$f(x) = \frac{x^2 - 3}{2x - 4}$$

a) vertical asymptote: $2x$-4 = 0, $x$ = 2.

b) Horizontal asymptote: **NONE** as it has a slant asymptote

# Examples



Find the asymptotes of the graphs of the functions

$$f(x) = \frac{x^2+1}{x^2-1}$$

(i) $\lim_{x \to 1^-} f(x) = -\infty$

Therefore the line $x = 1$

is a vertical asymptote.

(ii) $\lim_{x \to -1^-} f(x) = +\infty.$

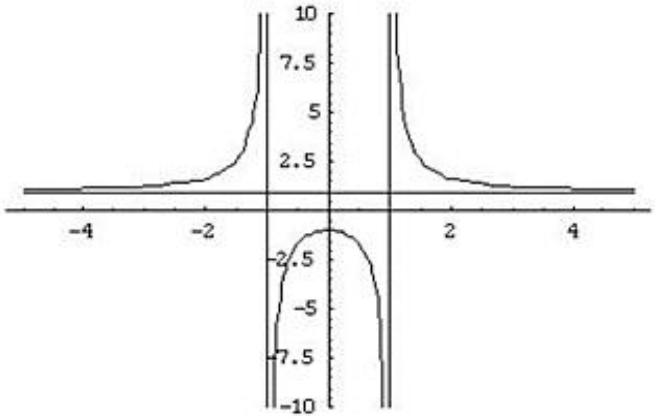
Therefore the line $x = -1$

is a vertical asymptote.

(iii) $\lim_{x \to \infty} f(x) = 1.$

Therefore the line $y = 1$

is a horizonatl asymptote.



# What you have learnt



- Understand one-sided limits
- Understand limits at infinity
- Seen limits at infinity of rational functions
- Understand horizontal asymptotes
- Understand what it means by infinite limits
- Understand vertical asymptotes

# Recollection



- We have seen:
  - Limits arise when finding the instantaneous rate of change of a function or the tangent to a curve
  - Informal definition of limit
  - How to calculate values of limit
  - Formal definition of limit
  - One-sided limits
  - Limit at infinity
  - Infinite limits
  - Horizontal and vertical asymptotes

# Continuity

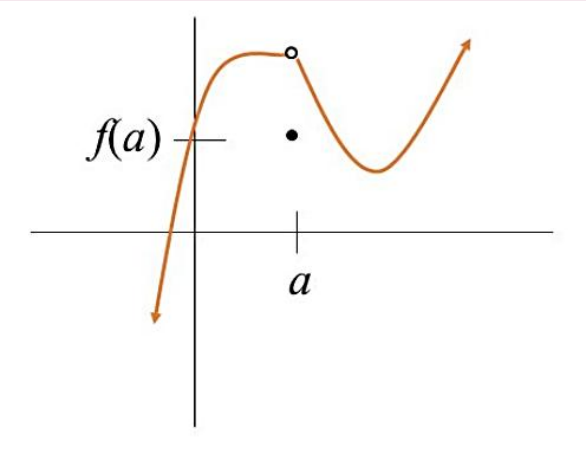


- Objectives:
  - To understand what it means for a function to be continuous
  - To study the properties of continuous functions
  - To be able to identify whether a function is continuous

# Continuity



- A function $f$ is *continuous* at the point $x = a$ if the following are true:
  - $f(a)$ is defined
  - $\lim_{x \to a} f(x)$ *exists*



# Definition of continuity



**DEFINITION**

*Interior point*: A function $y = f(x)$ is **continuous at an interior point *c*** of its domain if

$$\lim_{x \to c} f(x) = f(c).$$

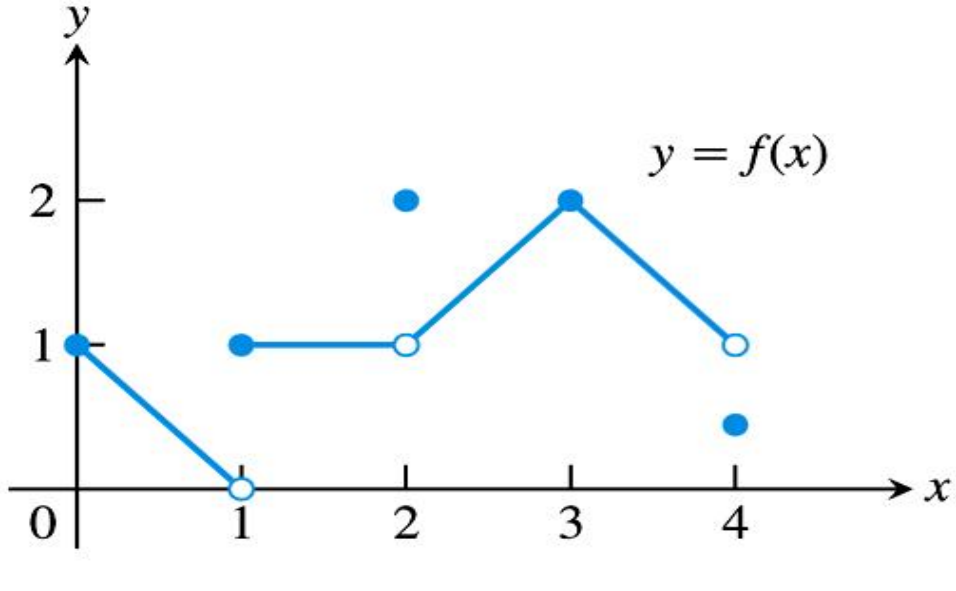
*Endpoint*: A function $y = f(x)$ is **continuous at a left endpoint *a*** or is **continuous at a right endpoint *b*** of its domain if

$$\lim_{x \to a^+} f(x) = f(a) \qquad \text{or} \qquad \lim_{x \to b^-} f(x) = f(b), \quad \text{respectively}.$$

# Continuity at a point



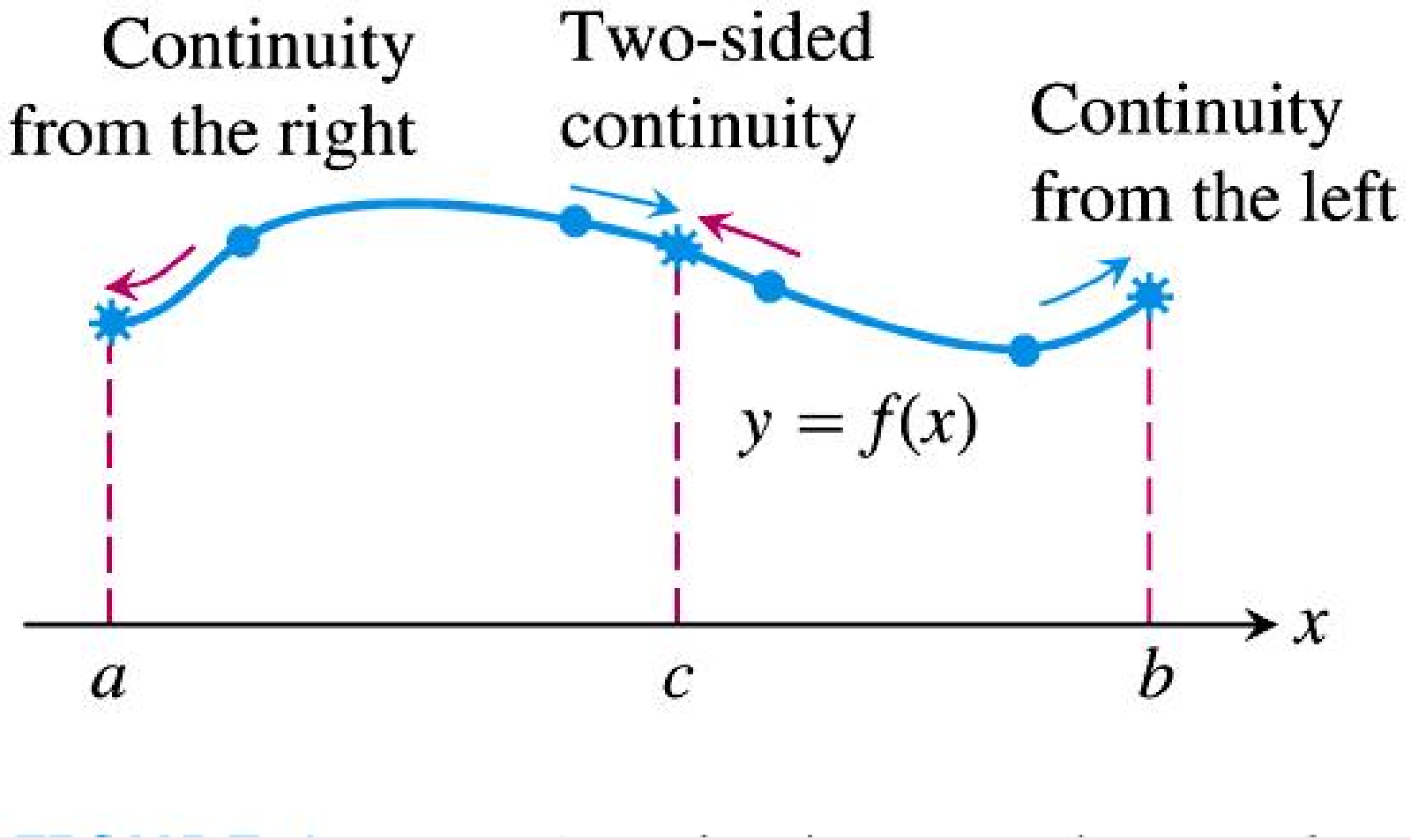
- Consider figure below. Find the points at which the function *f* is continuous and the points at which *f* is discontinuous.




- f(x) is discontinuous at x=1, 2 and 4. It is continuous at all other points of its domain.

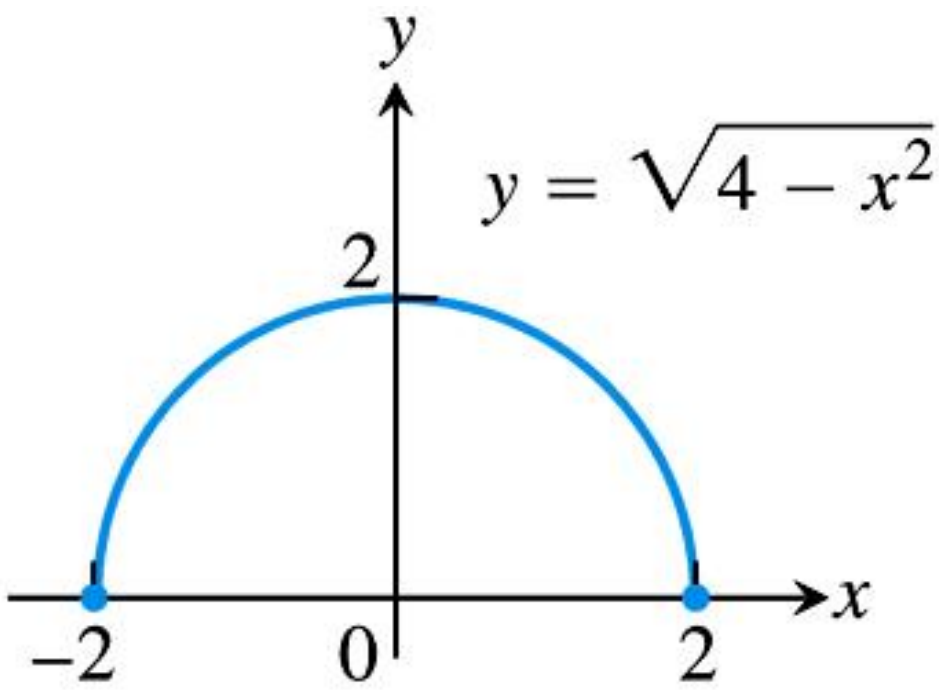
# Continuity at defined points





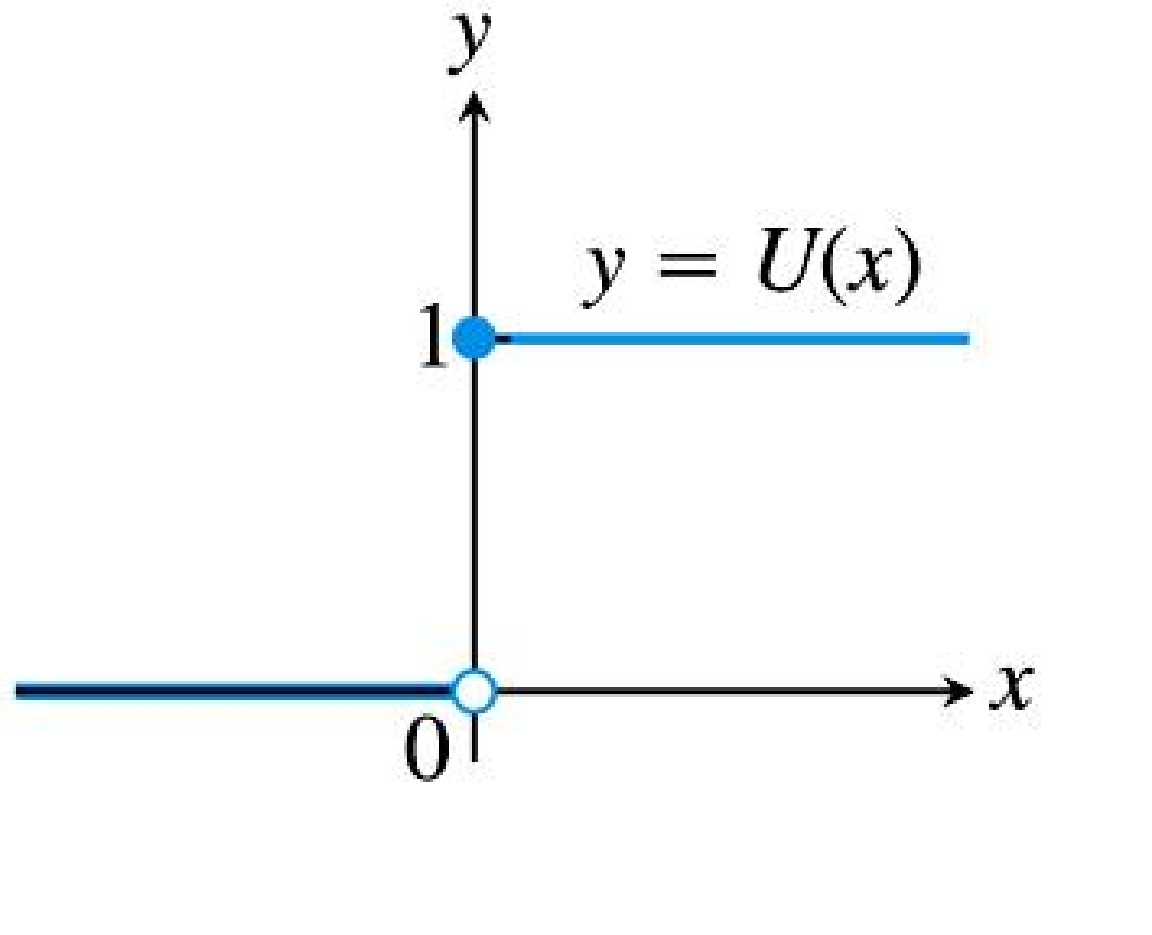
# Example 1





# Example 2





# Continuity Test



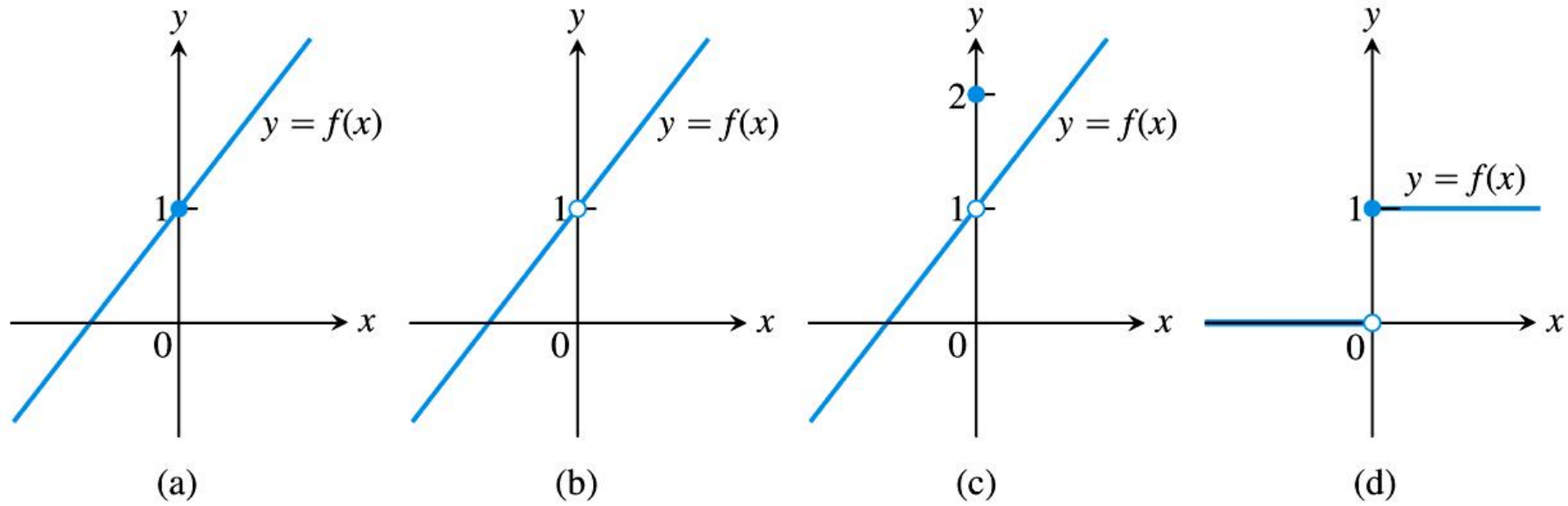
**Continuity Test**

A function $f(x)$ is continuous at $x = c$ if and only if it meets the following three conditions.

| | | |
|---|---|---|
| **1.** | $f(c)$ exists | ($c$ lies in the domain of $f$) |
| **2.** | $\lim_{x \to c} f(x)$ exists | ($f$ has a limit as $x \to c$) |
| **3.** | $\lim_{x \to c} f(x) = f(c)$ | (the limit equals the function value) |

# Example 3






a) Continuous at x = 0
b) Continuous at x = 0 if it had f(0) = 1
c) Continuous at x = 0 if it had f(0) = 1
d) Limit of f(x) does not exist – a jump discontinuity

# Continuous functions



- A function is continuous on an interval if and only if it is continuous at every point of the interval.
- A continuous function is one that is continuous at every point of its domain.

**THEOREM 9 Properties of Continuous Functions**

If the functions $f$ and $g$ are continuous at $x = c$, then the following combinations are continuous at $x = c$.

1. *Sums*: $f + g$
2. *Differences*: $f - g$
3. *Products*: $f \cdot g$
4. *Constant multiples*: $k \cdot f$, for any number $k$
5. *Quotients*: $f/g$ provided $g(c) \neq 0$
6. *Powers*: $f^{r/s}$, provided it is defined on an open interval containing $c$, where $r$ and $s$ are integers

# Example 4



Show that the rational function $f(x) = \frac{x+1}{x-2}$ is continuous at $x = 3$.

Steps: let $g(x) = x+1$, check its continuity at $x = 3$;

let $h(x) = x+2$, check its continuity at $x = 3$;

If both are continuous functions, then $f(x) = g(x)/ h(x)$ is also continuous; or else, vice versa.

# Continuous Functions

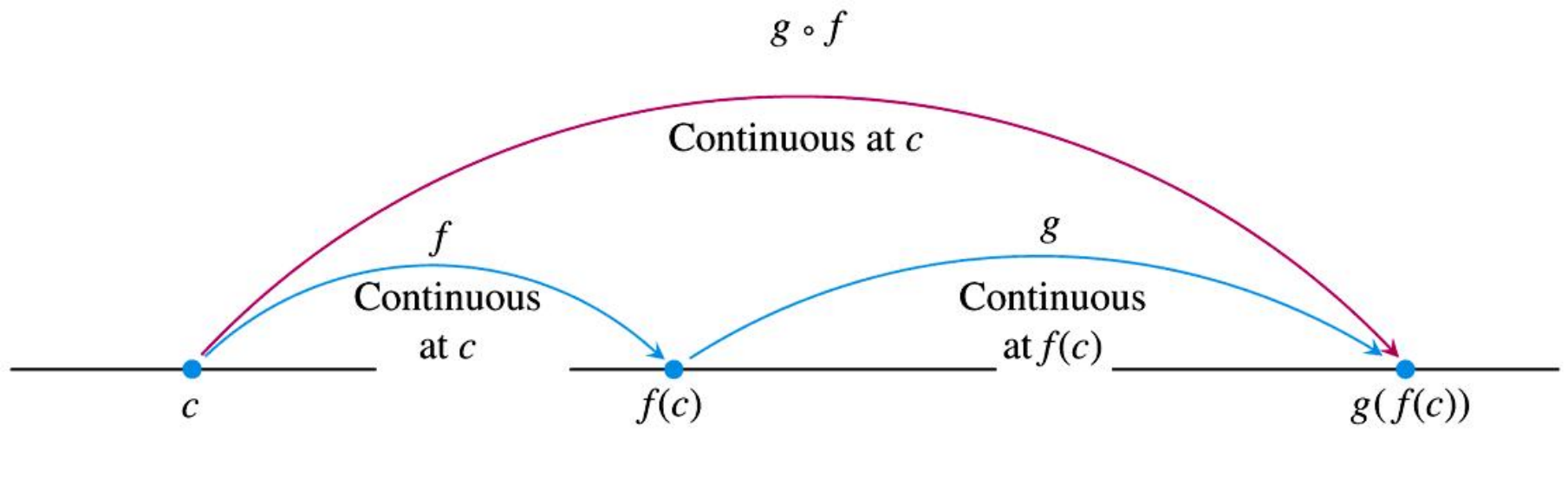


- Polynominal functions $f$ are continuous at every real number $c$ because $\lim_{x \to c}, f(x) = f(c)$.
- Rational functions are continuous at every point of their domains. They have points of discontinuity at the zeros of their denominators.
- The absolute value function $y = |x|$ is continuous at every real number.
- The exponential, logarithmic, trigonometric functions and radical function like $y = x^{(1/n)}$ ($n$ = a positive integer greater than 1) are continuous at every point of their domains.

# Composite of Continuous Functions



- All composite functions are continuous. The idea is that if $f(x)$ is continuous at $x=c$ and $g(x)$ is continuous at $x = f(c)$, then $g \circ f$ is continuous at $x = c$. Hence, the limit as $x \to c$ is $g(f(c))$.



# Theorems



**THEOREM 10 Composite of Continuous Functions**

If $f$ is continuous at $c$ and $g$ is continuous at $f(c)$, then the composite $g \circ f$ is continuous at $c$.

**THEOREM 11—Limits of Continuous Functions** If $g$ is continuous at the point $b$ and $\lim_{x \to c} f(x) = b$, then

$$\lim_{x \to c} g(f(x)) = g(b) = g(\lim_{x \to c} (f(x)).$$

# Example 5

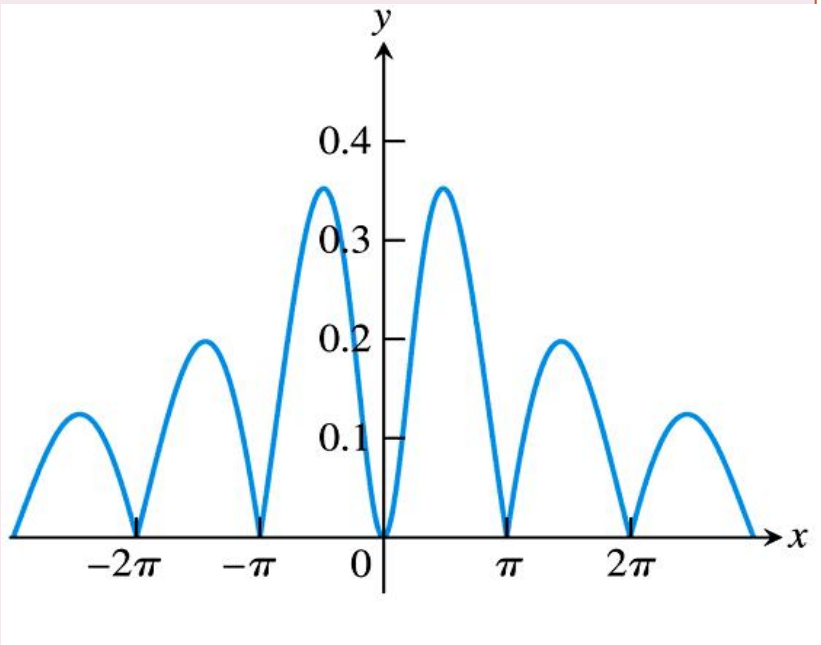


- Show that the following function is continuous everywhere on their respective domains.

$$y = \left| \frac{x \sin x}{x^2 + 2} \right|$$

- The function is continuous by

letting g(x) =|x|,f(x) = $\frac{x\sin x}{x^2+2}$



# Example 6



- Find the limits and determine if the function is continuous or not.

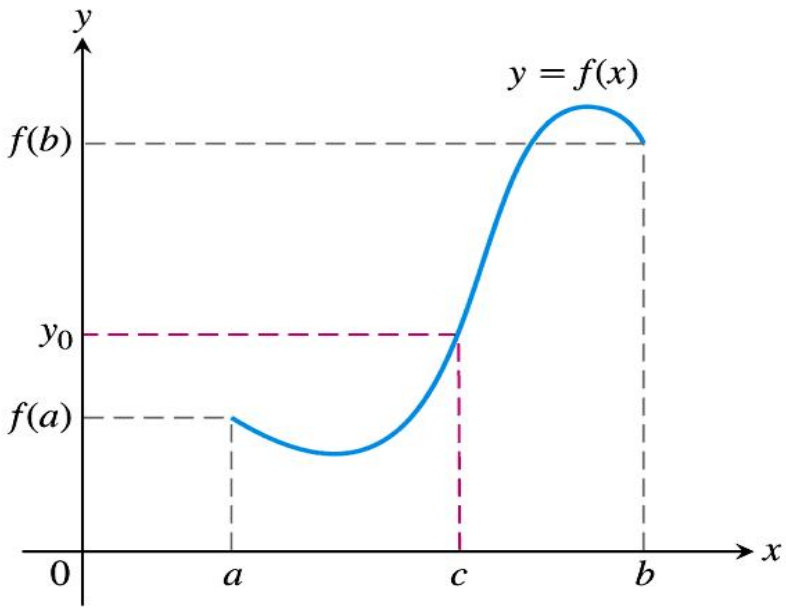
$$\lim_{x \to \pi} \sin(x - \sin x)$$

# Theorem



**THEOREM 12 The Intermediate Value Theorem for Continuous Functions**

A function $y = f(x)$ that is continuous on a closed interval $[a, b]$ takes on every value between $f(a)$ and $f(b)$. In other words, if $y_0$ is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some $c$ in $[a, b]$.



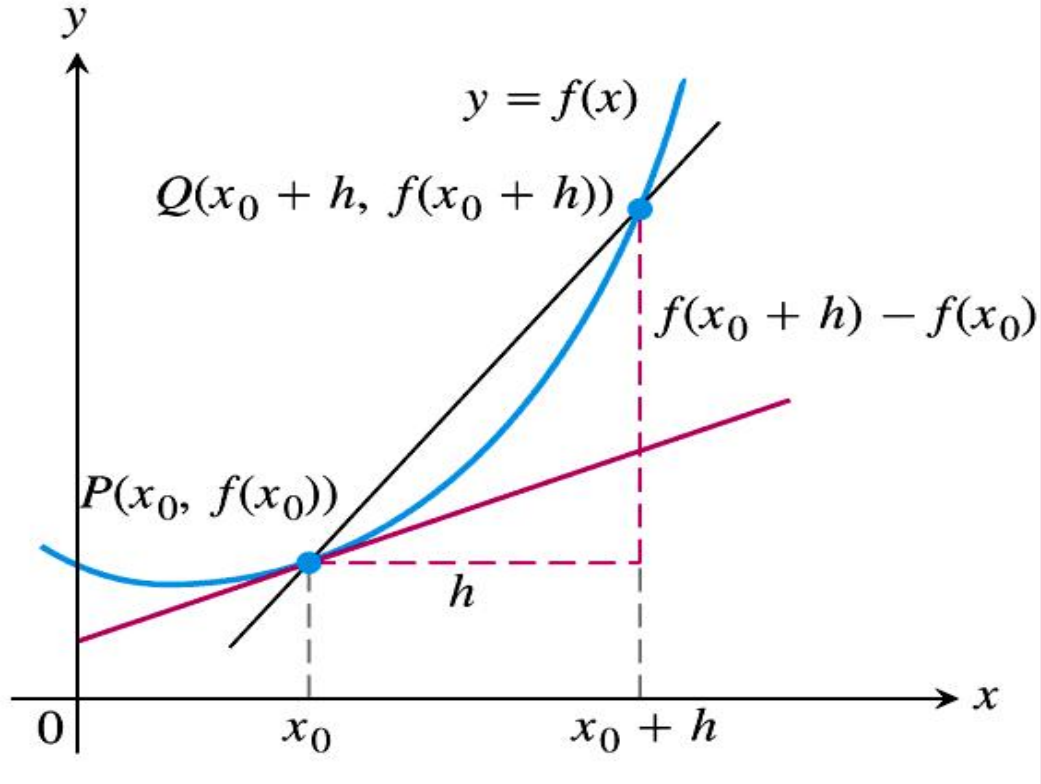
# Tangents and Derivatives at a Point



- Objectives
  - To be able to interpret the derivative of a function from the relationship between the notions of slopes and tangent lines to a curve to the instantaneous rate of change of a function

# Finding a Tangent to the Graph of a Function





# Definition of the slope of a curve



**DEFINITIONS** The **slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$m = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \qquad \text{(provided the limit exists).}$$

The **tangent line** to the curve at $P$ is the line through $P$ with this slope.

# Quick Exercise



Using the definition,

Find the slope of the curve $f(x) = \dfrac{1}{2+x}$ at $x = a$

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} = \lim_{h\to 0}\frac{\dfrac{1}{2+a+h}-\dfrac{1}{2+a}}{h}$$

$$= \lim_{h\to 0}\frac{1}{\mathrm{h}}\left(\frac{2+a}{(2+a+h)(2+a)}-\frac{2+a+h}{(2+a+h)(2+a)}\right)$$

$$= \lim_{h\to 0}\frac{1}{\mathrm{h}}\left(\frac{-h}{(2+a+h)(2+a)}\right)$$

$$= \lim_{h\to 0}\frac{-1}{(2+a+h)(2+a)}$$

$$= \frac{-1}{(2+a)^2}$$

# Rates of Change: Derivative at a Point



The expression

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

is called the difference quotient of $f$ at $x_0$ with increment $h$. If the difference quotient has a limit as $h$ approaches zero, that limit is given a special name and notation.

# Derivative of a function



**DEFINITION** The **derivative of a function $f$ at a point $x_0$**, denoted $f'(x_0)$, is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided this limit exists.

- Hence,

$$f'(x) = m$$

# Example



If air resistance is neglected, an object dropped from a great height will fall $s(t) = 16t^2$ feet in $t$ seconds. What is the object's velocity after $t = 2$ seconds?

$s(t) = 16t^2$, $\lim_{h \to 0} \frac{s(t+h) - s(t)}{h}$

at t $= 2$, $\frac{s(2+h) - s(2)}{h} = \frac{16(2+h)^2 - 16(2)^2}{h}$

$$= \frac{16(4 + 4h + h^2) - 16(2)^2}{h}$$

$$= \frac{64h + 16h^2}{h}$$

$$= 64 + 16h$$

$\therefore \lim_{h \to 0} 64 + 16h = 64$.

# Brief notes



- Average speed = $\frac{distance}{time}$
- Average rate of change of $f(x)$ wrt $x$ for a function $f$ as $x$ changes from $x1$ to $x2$: $\frac{f(x_2)-f(x_1)}{x_2-x_1}$
- Slope = $\frac{change\ in\ y}{change\ in\ x} = \frac{change\ in\ distance}{change\ in\ time}$ = average speed
- Slope of secant line : $\frac{\Delta y}{\Delta x} = \frac{f(x_1+h)-f(x_1)}{h}$
- Instantaneous rate of change: $\lim_{h\to 0}\frac{f(x_1+h)-f(x_1)}{h}$
- Slope of tangent line: $\lim_{h\to 0}\frac{f(x_1+h)-f(x_1)}{h}$
- The derivative of the function $f$ at $x$ is defined as:

$$f'(x) = \lim_{h\to 0}\frac{f(x_1+h)-f(x_1)}{h}$$

# Summary of Chapter 2



- We learned about:
  - Slopes of curves
  - Lines tangent to a curve
  - The rate of change of a function
  - Limits and continuity
  - Derivative of a function at a point

1. The slope of $y = f(x)$ at $x = x_0$
2. The slope of the tangent to the curve $y = f(x)$ at $x = x_0$
3. The rate of change of $f(x)$ with respect to $x$ at $x = x_0$
4. The derivative of $f$ at $x = x_0$
5. The limit of the difference quotient, $\lim_{h \to 0} \dfrac{f(x_0 + h) - f(x_0)}{h}$